

PROBLEM SHEET 6

This problem sheet is due in on Gradescope by **9am** on **Monday 28 February 2022**.

Topics covered: Chapters 3-4

1. **Turing Analysis of a Type III Functional Response Predator-Prey Model**

Consider the predator-prey model,

$$\begin{aligned}\frac{\partial u}{\partial t} &= u(1-u) - \frac{avu^2}{1+bu^2} + \frac{\partial^2 u}{\partial x^2} \\ \frac{\partial v}{\partial t} &= -cv + \frac{avu^2}{1+bu^2} + D \frac{\partial^2 v}{\partial x^2},\end{aligned}\tag{1}$$

where all parameters are positive, and we have chosen a timescale such that D represents the ratio of diffusion parameters. We consider the spatial domain $x \in [0, L]$ with Neumann boundary conditions.

- (a) Find the three biologically feasible spatially homogeneous equilibria of this model. Hint: First consider equilibria with $v^* = 0$, and then any which satisfy $v^* \neq 0$.

SOLUTION: The equilibria satisfy:

$$\begin{aligned}0 &= u^*(1-u^*) - \frac{av^*(u^*)^2}{1+b(u^*)^2} \\ 0 &= -cv^* + \frac{av^*(u^*)^2}{1+b(u^*)^2}.\end{aligned}\tag{2}$$

If we set $v^* = 0$, then either $u^* = 0$ or $u^* = 1$, so two of the equilibria are given by $(0, 0)$ and $(1, 0)$.

For the third equilibrium, we can divide the first equation by u^* and solve for v^* to find,

$$v^* = \frac{(1-u^*)(1+b(u^*)^2)}{au^*}.$$

Alternatively: We can subtract the second equation from the first to find:

$$u^*(1-u^*) - cv^* = 0 \implies v^* = \frac{u^*(1-u^*)}{c}.$$

These expressions appear different but at the steady state we find below for u^* , turn out to give the same value for v^* .

To find u^* , we could substitute either of these expressions into the second equation and rearrange, or we can instead recognize that since we assume $v^* \neq 0$, we can divide the second equation by v^* to find,

$$-c + \frac{a(u^*)^2}{1+b(u^*)^2} = 0 \implies (a-cb)(u^*)^2 - c = 0 \implies u^* = \sqrt{\frac{c}{a-cb}},$$

where we have only considered the positive root as the negative solution is not biologically feasible. Additionally, we need $a > cb$ for this state to be feasible. We also need $u^* < 1$ for v^* to be positive, which implies $a - cb > c$ so $a > c(1 + b)$, which is a stricter condition than the one found above.

- (b) Which of these steady states does it make sense to consider for a diffusion-driven instability? Hint: Consider what an initial inhomogeneous perturbation would look like for each case, and determine if any of these are physically unrealistic.

SOLUTION: For any steady state with $v^* = 0$, spatial oscillations would lead to non-physical values of v^* , and hence only the third equilibrium would make sense for a diffusion-driven instability.

- (c) For the equilibrium which can in principle exhibit a diffusion-driven instability, compute the Jacobian matrix and determine conditions for this equilibrium to be stable in the absence of diffusion. Hint: Before plugging in the values for this steady state, can you immediately spot simplifications of any terms in the Jacobian matrix?

SOLUTION: The Jacobian matrix is:

$$J(u^*, v^*) = \begin{pmatrix} 1 - 2u^* - \frac{2av^*u^*}{(1+b(u^*)^2)^2} & -\frac{a(u^*)^2}{1+b(u^*)^2} \\ \frac{2av^*u^*}{(1+b(u^*)^2)^2} & -c + \frac{a(u^*)^2}{1+b(u^*)^2} \end{pmatrix}.$$

While this looks quite grisly, we can see that the bottom-right component is exactly the equation we had for u^* , and hence must be 0. Hence we actually have that $\det(J) > 0$ without computing any of the terms, as the off-diagonal terms are negative and positive respectively. That is,

$$J(u^*, v^*) = \begin{pmatrix} 1 - 2u^* - \frac{2av^*u^*}{(1+b(u^*)^2)^2} & -\frac{a(u^*)^2}{1+b(u^*)^2} \\ \frac{2av^*u^*}{(1+b(u^*)^2)^2} & 0 \end{pmatrix} \implies \det(J) = \frac{2a^2(u^*)^3v^*}{(1+b(u^*)^2)^3} > 0.$$

To satisfy the condition $\text{tr}(J) < 0$, we must have

$$1 < 2u^* + \frac{2av^*u^*}{(1+b(u^*)^2)^2} = 2u^* + 2(1 - u^*) = 2,$$

which is also automatically satisfied, as long as this steady state is feasible (that is, $a > cb$).

- (d) Can any equilibria for this system exhibit a Turing instability? Why or why not?

SOLUTION: As the other two equilibria are unphysical to consider Turing instabilities around, we only need to consider the third equilibrium analyzed above. In this case, as the bottom-right component of the Jacobian is zero, we immediately have that this system cannot exhibit a Turing instability. The

requirements of the sign structure of the Jacobian are strict positive or negative values at the equilibrium. So a zero means that if the homogeneous equilibrium is stable in the absence of diffusion, it must remain stable even with diffusion. Hence, none of these equilibria can exhibit a Turing instability.

2. Non-Euclidean 3D Patterns

Consider the eigen-equation

$$\nabla^2 w_k + \rho_k w_k = 0, \quad (3)$$

in a three dimensional cylindrical space (r, θ, z) , with $r \in [0, L_r]$ the radial coordinate, $\theta \in [0, 2\pi)$ the angular coordinate and $z \in [0, L_z]$ the vertical coordinate. In this coordinate system the Laplacian takes the form

$$\nabla^2 w_k = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial w_k}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 w_k}{\partial \theta^2} + \frac{\partial^2 w_k}{\partial z^2}. \quad (4)$$

We assume the eigenfunctions satisfy Neumann conditions at the top, bottom, and outer boundaries of the cylinder, and by continuity in θ must be periodic. That is,

$$\begin{aligned} \frac{\partial w_k}{\partial z}(r, \theta, 0) &= \frac{\partial w_k}{\partial z}(r, \theta, L_z) = \frac{\partial w_k}{\partial r}(L_r, \theta, z) = 0, \\ w_k(r, 0, z) &= w_k(r, 2\pi, z), \quad \frac{\partial w_k}{\partial \theta}(r, 0, z) = \frac{\partial w_k}{\partial \theta}(r, 2\pi, z). \end{aligned} \quad (5)$$

- (a) Consider a separable solution in the form $w_k = R(r)P(\theta)Z(z)$ (the P implying periodic). Substitute this into (3) and show that

$$\frac{1}{Rr} \frac{dR}{dr} + \frac{1}{R} \frac{d^2 R}{dr^2} + \frac{1}{Pr^2} \frac{d^2 P}{d\theta^2} + \rho_k = -\frac{1}{Z} \frac{d^2 Z}{dz^2}. \quad (6)$$

SOLUTION: On substituting (and dropping the variable dependencies so $w_k = RPZ$) we have,

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial (RPZ)}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 (RPZ)}{\partial \theta^2} + \frac{\partial^2 (RPZ)}{\partial z^2} + \rho_k RPZ = 0.$$

Dividing this through by $w_k = RPZ$, and taking the derivatives through the functions which do not depend on those variables, we have

$$\frac{1}{Rr} \frac{d}{dr} \left(r \frac{dR}{dr} \right) + \frac{1}{Pr^2} \frac{d^2 P}{d\theta^2} + \frac{1}{Z} \frac{d^2 Z}{dz^2} + \rho_k = 0,$$

where we have also replaced the partial derivatives by total ones, as these functions only depend on a single variable each. Finally we can move the Z term to the other side of the equation, and carry out the product rule for the r derivatives, to obtain (6).

- (b) We now set both the right hand side and left hand side of (6) equal to some constant λ , i.e.

$$-\frac{1}{Z} \frac{d^2 Z}{dz^2} = \lambda, \quad (7)$$

$$\frac{1}{Rr} \frac{dR}{dr} + \frac{1}{R} \frac{d^2 R}{dr^2} + \frac{1}{Pr^2} \frac{d^2 P}{d\theta^2} + \rho_k = \lambda. \quad (8)$$

Show that, depending on the sign of λ , the solutions for Z are either.

$$Z_\lambda(z) = C_1 e^{\sqrt{|\lambda|}z} + C_2 e^{-\sqrt{|\lambda|}z}, \quad (9)$$

$$Z_\lambda(z) = C_3 z + C_4, \quad (10)$$

or

$$Z_\lambda(z) = C_5 \cos(\sqrt{\lambda}z) + C_6 \sin(\sqrt{\lambda}z), \quad (11)$$

where the C_i are constants of integration. Which of these satisfies the boundary conditions (5)? What does this mean in terms of the permissible sign(s) and values of λ ?

SOLUTION: We can write the equation for Z as,

$$Z'' + \lambda Z = 0.$$

This is a constant-coefficient ODE, and so we can solve it by setting $Z = \exp(\mu z)$ to find that $\mu^2 = -\lambda$. If $\lambda < 0$, then we have solutions of the exponential form (9). If $\lambda = 0$, we have solutions of the form (10). Finally if $\lambda > 0$, we have solutions of the form (11).

As in the Lecture Notes, we can go through each case and show that the first two cannot satisfy the boundary conditions. For the third, we find that the coefficient of $\sin$ must be zero, and that λ must satisfy:

$$C_6 \cos(\sqrt{\lambda}L_z) = 0 \implies \lambda = \left(\frac{\pi n_z}{L_z}\right)^2, \quad n_z = 0, 1, 2, \dots$$

Hence these are the permissible, positive values of λ .

- (c) Show that the remaining r, θ equation can be written as

$$-\frac{1}{P} \frac{d^2 P}{d\theta^2} = \kappa, \quad \frac{r}{R} \frac{dR}{dr} + \frac{r^2}{R} \frac{d^2 R}{dr^2} + (\rho_k - \lambda)r^2 = \kappa, \quad (12)$$

where κ is another unknown constant.

SOLUTION: Multiplying (7) by r^2 , and swapping the λ and P terms, we have

$$\frac{r}{R} \frac{dR}{dr} + \frac{r^2}{R} \frac{d^2 R}{dr^2} + (\rho_k - \lambda)r^2 = -\frac{1}{P} \frac{d^2 P}{d\theta^2} = \kappa,$$

where these both equal κ as the left side is only a function of r , and the right only a function of θ .

(d) Explain why

$$P(\theta) = \sum_{n=0}^{\infty} A \cos(n\theta) + B \sin(n\theta) \quad (13)$$

and hence why κ is a *positive* integer. Compute the permissible values of κ .

SOLUTION: We can use the solution approach in part (b) to see that P takes one of the forms given in (9)-(11). As before, checking the boundary conditions we see only the sinusoidal functions can satisfy these. Unlike in that case, however, we cannot eliminate either function. We can compute κ by considering the boundary conditions in detail. We have:

$$A \cos(\sqrt{\kappa}0) + B \sin(\sqrt{\kappa}0) = A = A \cos(\sqrt{\kappa}2\pi) + B \sin(\sqrt{\kappa}2\pi),$$

and

$$-A\sqrt{\kappa} \sin(\sqrt{\kappa}0) + B\sqrt{\kappa} \cos(\sqrt{\kappa}0) = B\sqrt{\kappa} = A\sqrt{\kappa} \sin(\sqrt{\kappa}2\pi) + B\sqrt{\kappa} \cos(\sqrt{\kappa}2\pi).$$

In each case we must choose κ such that $\sin(\sqrt{\kappa}2\pi) = 0$ which requires $\sqrt{\kappa}2\pi = n\pi$ so that $\sqrt{\kappa} = 0, n/2, n, 3n/2, \dots$. However, we also require $\cos(\sqrt{\kappa}2\pi) = 1$, and hence the half-integer values of $\sqrt{\kappa}$ are inadmissible. So we have that $\kappa = n^2$ with $n = 0, 1, 2, \dots$, which is precisely what we have in formula (13).

(e) **Optional** Show by substituting $x = \sqrt{\rho_k - \lambda}r$ and writing $n_\theta^2 = \kappa$ that the equation for R , given by the second of (12), becomes

$$\frac{d^2 R}{dx^2} + \frac{1}{x} \frac{dR}{dx} + \left(1 - \frac{n_\theta^2}{x^2}\right) R = 0, \quad (14)$$

the Bessel equation we have seen in class. What are the solutions to this equation?

SOLUTION: We have that the derivatives transform as,

$$\frac{d}{dr} = \sqrt{\rho_k - \lambda} \frac{d}{dx}, \quad \frac{d^2}{dr^2} = (\rho_k - \lambda) \frac{d^2}{dx^2}.$$

So Equation (12) becomes:

$$\frac{1}{R} \frac{dR}{dx} + \frac{x^2}{R} \frac{d^2 R}{dx^2} + x^2 = n_\theta^2.$$

Multiplying through by R/x^2 , we have the result.

The solutions are given by Bessel functions of the first kind, $J_{n_\theta}(x)$. So $R(r) = J_{n_\theta}(r/\sqrt{\rho_k - \lambda})$.

(f) **Optional** State the full solution w_k (you are welcome to quote formulas from the lecture notes involving the Bessel equation or otherwise).

SOLUTION: Putting everything together, we have that Z is given by $\cos(\pi n_z z / L_z)$,

with $\lambda = (\pi n_z/L_z)^2$, P is given by (13) with $\kappa = n^2$, and $R = J_{n_\theta}(r/\sqrt{\rho_k - \lambda})$. Hence a full solution is given by:

$$w_k = w_{(n_r, n_\theta, n_z)} = \sum_{n_r=0}^{\infty} \sum_{n_\theta=0}^{\infty} \sum_{n_z=0}^{\infty} (A_{(n_r, n_\theta, n_z)} \cos(n_\theta z) + B_{(n_r, n_\theta, n_z)} \sin(n_\theta z)) \cos\left(\frac{\pi n_z z}{L_z}\right) J_{n_\theta}\left(\frac{r}{\sqrt{\rho_k - \frac{\pi n_z^2}{L_z^2}}}\right).$$

Note that we have not computed ρ_k , and that n_r does not appear in our formula! This can be done by solving for the roots of the derivative of the bessel function at $r = R$, so that the full spatial eigenvalue ρ_k will really be a function of all three mode numbers. However, solving for these roots must be done numerically (or approximately for ‘large’ or ‘small’ roots), but this is beyond the scope of our class.
